



Data Driven Education

Natural Language Processing (NLP) Subteam
Presentation 3 - 4/15/2020

Nine Sub-meetings: 2/7 2/14 2/21 2/28

3/6 3/13 3/31

4/3 4/10

Team Members



Yuntian Zhang

Project Manager
Data Scientist



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Web Developer

Researcher



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Researcher



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Data Science Team

Team Focus:

- edX data
 - Shifted focus from Piazza data to edX data
 - 2 tables: forum_thread_ferpa & forum_comment_ferpa
 - 2667 comments in total: includes online classes such as CS1301, ISYE6501, etc
 - Focus on CS1301-Online - 2137 comments in total
- Cognitive presence
 - Four phases of cognitive presence
 - Created a document on categorizing cognitive presense
 - Labeled 500+ comments for model training
 - Next step: build a automatic categorizing model (Still undergoing...)
- NLP analysis
 - Basic statical analysis on different phases
 - Overall NLP analysis on each phase of cognitive presence
 - Word frequency and word cloud analysis

Data Science Team

- edX data

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id	thread_id	parent_id	user_id	course_id	body	CP Code	anonymous	anonymous	upvotes	downvotes	abuse_flag	historical_a
\x3563653	\x35633666366366633		23482927	course-v1:C	I know that probably you have solved it by now, but just in case if someone else looking at this problem. My solution is not optimized and I am sure there are a lot simpler solutions, but it works. ![solution] [1] [1]: https://edxuploads.s3.amazonaws.com/15582729273110228.png	3	FALSE	FALSE	0	0	0	0
\x3539336	\x3539336	\x3539336	14429895	course-v1:C	I gather, after reading [here][1] and playing around, that float numbers are calculated to 17 significant digits, and since 62.364×298 can't be represented accurately, there's a small error in the result, which Python shows, unless we format the result e.g. <code>format(62.364*298, '4f')</code> in the problem question we're talking of, the result of 62.364×298 is <code>18584.471999999998</code> { try it : <code>print(62.364*298)</code> } but it so happens that 10×62.364 has no error, and nor does $10 \times 62.364 \times 298$ (Python multiplies left to right... but $10 \times (62.364 \times 298)$ has the same error taking different numbers : $397 \times 62.364 = 24758.507999999998$ but if we multiply by ten either before or after, and regardless of brackets, the result is correct. [1]: https://docs.python.org/3/tutorial/floatpoint.html	2	FALSE	FALSE	0	0	0	0

Data Science Team

- Cognitive presence
 - Four phases of cognitive presence
 - Created a document on categorizing cognitive presence
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0: Non-cognitive presence

- comments that are not clear (e.g., picture and code w/o comment)
- thank you and socializing comments
- logistics, deadlines, technical Q&As

1: Triggering Event

- disagree/conflict with prior knowledge. "I think this is right/wrong." "I'm stuck." "I have no idea." "I am clueless."
- focus on problem itself
- shorter than 2 sentences in general – just the question, no elaboration
- do not show they understand the concepts or ways that are needed to solve the given problem

2: Exploration

- more explorative, some hesitation, asking or giving suggestions, less authority ("I am not a TA, but ...")
- show that they understand something, but don't understand something else – having partial understanding ("I understand this but not that")
- introduce new ideas or suggestion ("Maybe try this?")

3: Integration

- give someone solution
- use textbook references or other credible sources to help find solutions
- "[their solution]. This works for me."

4: Resolution

- include elaboration on the solution;
- whenever comment is definite/sure.
- "tested code". "it worked"
Defending solutions

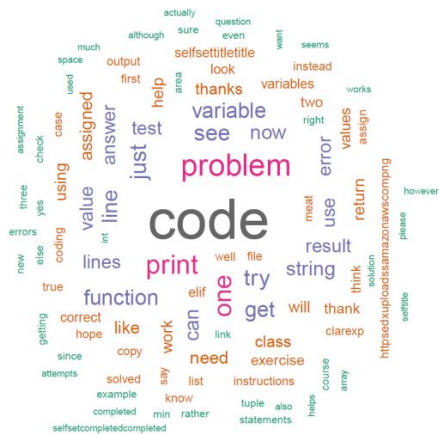
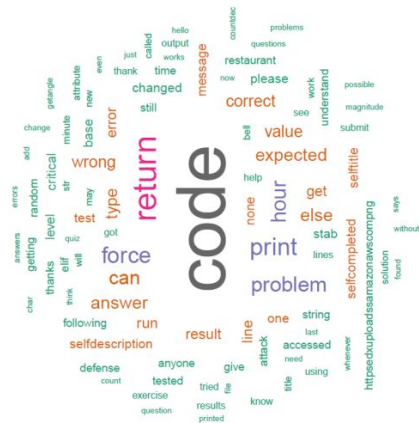
Data Science Team

- NLP analysis
 - Basic statical analysis on different phases
 - overall NLP analysis on each phases
 - Word frequency and word cloud analysis

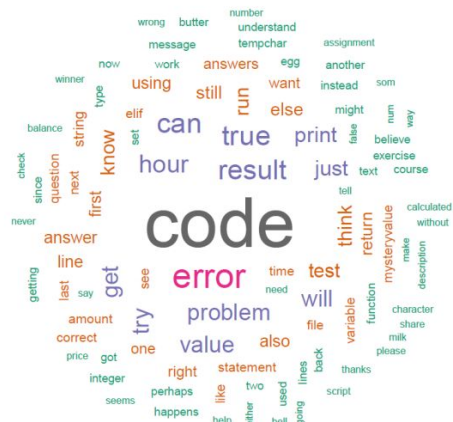
	count	average_polarity	average_subjectivity
overall	541.0	0.063934	0.337881
0	189.0	0.128996	0.310097
1	57.0	-0.034934	0.404077
2	104.0	0.032042	0.340136
3	106.0	0.069096	0.339052
4	85.0	0.018150	0.351047

Phase 1

Triggering Event

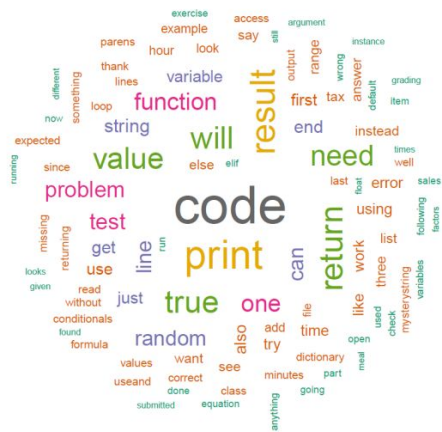


Phase 3 Integration



Phase 2

Exploration



Phase 4 Resolution

Word Cloud Data Frame: Top 25 Words & Frequency

Social/Teaching Presence

word	freq
course	48
thanks	30
thank	26
code	23
see	20
can	20
python	18
problem	17
return	16
great	14
work	14
httpsedu	
ploadssa	
mazonaw	
scompng	14
everyone	13
else	13
learn	12
got	12
program	
ming	12
like	12
will	12
well	12
hey	12
name	11
get	11
help	11
one	11

Cognitive Presence

word	freq
code	246
print	99
return	89
problem	81
result	79
can	73
value	72
error	65
will	62
TRUE	60
get	58
one	58
line	57
test	57
hour	55
answer	52
just	49
else	48
string	46
try	46
need	45
function	43
see	40
correct	39
variable	39

Cognitive Presence

Phase 1

word	freq
code	94
return	40
print	30
force	27
problem	24
hour	24
can	21
answer	20
else	20
expected	19
wrong	18
value	18
type	17
correct	16
line	15
run	15
get	14
result	14
error	14
none	13
message	13
one	12
test	12
selfcomple	
ted	12
selfdescrip	
tion	12

Phase 2

word	freq
code	63
error	29
result	22
TRUE	22
can	22
hour	20
problem	18
get	17
value	17
try	17
will	17
just	16
print	16
test	15
know	14
think	14
run	14
still	13
else	13
using	12
also	12
first	11
answer	11
return	11
line	11

Phase 3

word	freq
code	40
problem	20
print	17
one	16
try	14
just	14
see	14
line	13
function	13
get	13
variable	13
can	12
now	12
string	12
test	11
answer	11
result	11
use	11
error	11
value	11
lines	11
assigned	10
like	10
need	10
class	9

Phase 4

word	freq
code	49
print	36
result	32
return	29
will	28
TRUE	28
value	26
need	25
one	20
function	20
problem	19
test	19
line	18
can	18
random	18
string	15
end	14
get	14
just	13
variable	13
also	12
first	12
use	12
try	11
time	11

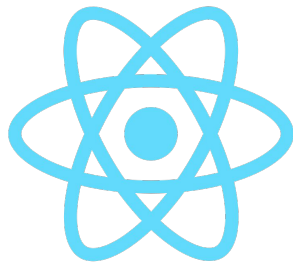
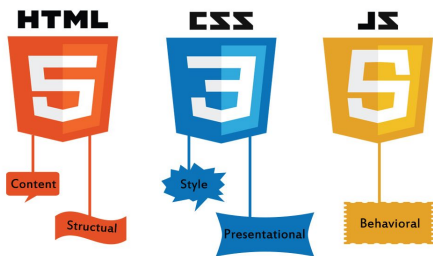
Data Science Team

- Future job
 - Keep building the model for automatic categorizing
 - Generalize the model & Categorize document
 - More NLP analysis (using NLTK, etc)
 - Visualizations on the results
 - More integration with the web-app development team

Special Thanks to Jonna for her professional help!!!

Web Application Development Team

- Became familiar with Flask, HTML, CSS, Javascript, React, and SQL
- Established the file/folder organization for the web app
- Designed UI mockups
- Developed a preliminary web app
- Created a unified design document



Mockups - Modular

Some Cool Title

Search

Choose a class:

CS 1301	INTA 4060	PHIL 3103	SPAN 1001
CS 1331	LMC 2500	PHIL 4174	SPAN 1002
CS 1332	LMC 2720	PHYS 2211	SPAN 1813
CS 2261	LMC 3112	PHYS 2212	
CS 2340	LMC 3214	PHYS 3043	
CS 3450	LMC 3216	PHYS 3122	
CS 4460	LMC 3262	PHYS 3211	
ENGL 1101	MATH 1551	PHYS 6104	
ENGL 1102	MATH 1552	PHYS 6202	
INTA 1110	MATH 1553	RUSS 1001	
INTA 2040	MATH 2550	RUSS 1002	
INTA 2210	MATH 3012	SLS 3110	
INTA 3012	PHIL 2010	SLS 3120	
INTA 3110	PHIL 3102	SOC 1101	

Some Cool Title

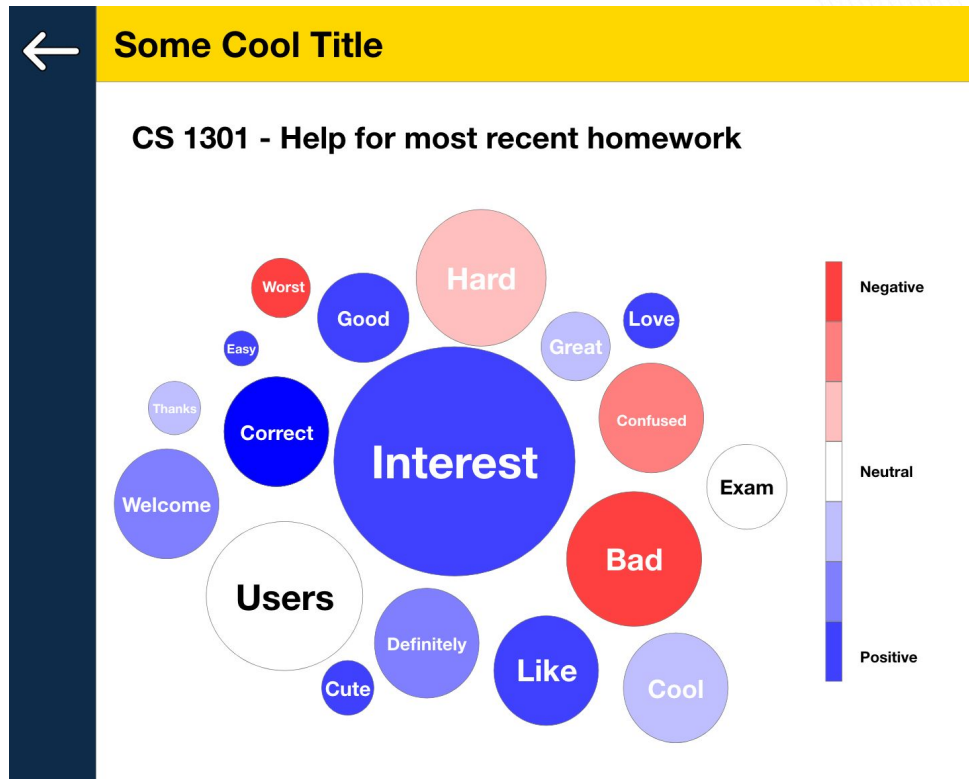
Search

CS 1301

Sort: Oldest to Newest

Choose a discussion post:

- [Help for most recent homework](#)
- [Yellow Jackets vs Dawgs](#)
- [No Title](#)
- [hello everyone welcome](#)
- [is this an announcement?](#)
- [I'm confused...](#)
- [today was a bad day](#)
- [how was the exam?](#)
- [today was a good day](#)
- [Combo is the worst class](#)
- [My Topic](#)

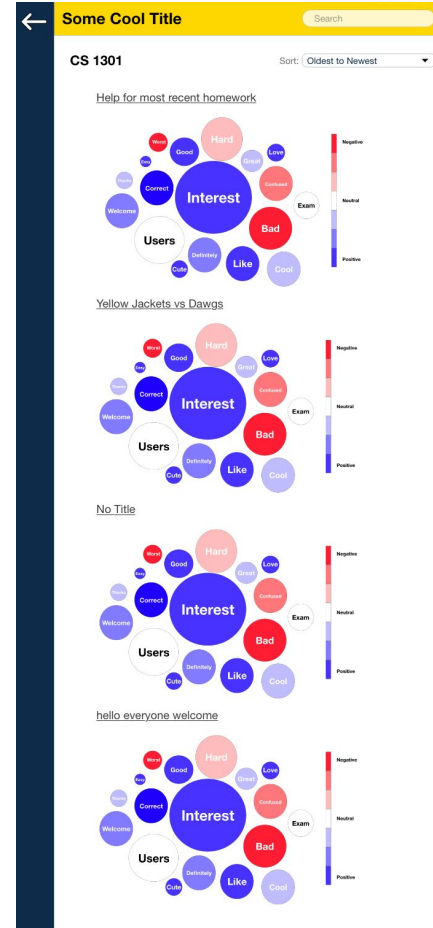


Mockups - Data Dump

Some Cool Title

Choose a class:

CS 1301	INTA 4060	PHIL 3103	SPAN 1001
CS 1331	LMC 2500	PHIL 4174	SPAN 1002
CS 1332	LMC 2720	PHYS 2211	SPAN 1813
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Preliminary Web App

Courses

- sstaudaher3 Sandbox 20190716

Discussion	Sentiment
I'm going to use this discussion forum to force my cat upon all of you, here he is.	Sentiment(polarity=0.0, subjectivity=0.0)
:-)	Sentiment(polarity=0.5, subjectivity=1.0)

Discussion	Sentiment
I am currently taking combinatorics and it is the worst class I've ever taken.	Sentiment(polarity=-0.5, subjectivity=0.7)
It is extremely hard and a ton of work.	Sentiment(polarity=-0.2916666666666667, subjectivity=0.5416666666666666)
I much prefer this class!	Sentiment(polarity=0.25, subjectivity=0.2)

Discussion	Sentiment
i think it was pretty easy	Sentiment(polarity=0.3416666666666667, subjectivity=0.9166666666666667)

Discussion	Sentiment
lets find out	Sentiment(polarity=0.0, subjectivity=0.0)

Discussion	Sentiment
i failed my exam 1. homework 1 is due tonight but i haven't started yet.	Sentiment(polarity=-0.3125, subjectivity=0.3375)
this class is so hard.	Sentiment(polarity=-0.2916666666666667, subjectivity=0.5416666666666666)

Discussion	Sentiment
bc it rained all day!	Sentiment(polarity=0.0, subjectivity=0.0)
i like rains!	Sentiment(polarity=0.0, subjectivity=0.0)

Discussion	Sentiment
yay	Sentiment(polarity=0.0, subjectivity=0.0)

Web Application Development Team Goals for Fall 2020

- Continue development of web application based on the results of the survey
- Incorporate Dash for data visualization
- Construct a relational database for discussion data

Education Research: Gathering Opinions

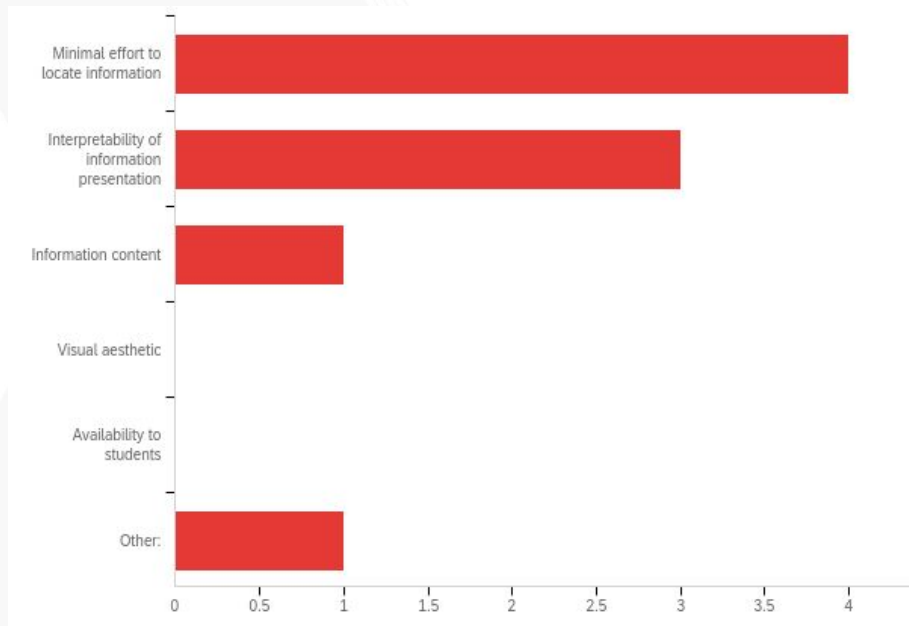
- **Goal:** determine usefulness and possible improvements for Canvas web app
- **Audience:** Professors and TAs in CS, Psychology, Math, and other disciplines
- **Method:** Qualtrics survey with app mockups
- **Obstacles:** Later launch and lower participation than intended due to moving online

Education Research: Results

- 4 responses
- 3 out of 4 responded positively to Navigable format
- 2 out of 4 responded positively to Waterfall format
- 2 out of 4 said they could see using the app in the classroom
- General Complaints:
 - Use of bubbles and visualizations were unclear
 - Not capturing use case of students using forums
 - Don't like extra complications

Education Research: Results

Importance of Factors

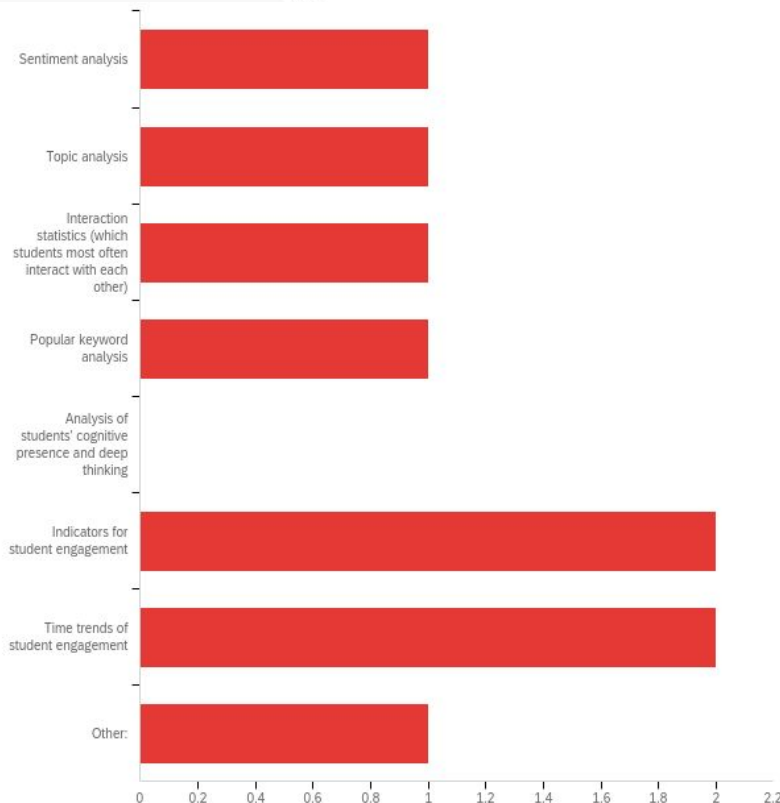


- Most important: accessibility and interpretability of information
- Not important: aesthetic and student accessibility
- Other: “ease of installation into Canvas

Education Research: Results

Importance of Features

- Most important: student engagement analysis
- Not important: Cognitive Presence and Deep Thinking



Education Research: Moving Forward

- Collect more data!
- For future surveys:
 - Make it clear that filler text is, in fact, filler text
 - Explain the included visualizations more clearly
 - Emphasize app as a statistics/ course information aid, not a discussion tool itself
- Development Recommendations:
 - Tentative focus on navigable format over waterfall
 - Look into Student Engagement metrics

Questions

?